Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/30/2010
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism in Micro- and Macro Scales		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Didion, Jeffrey (NASA Goddard Space Flight Center) Cassel, Kevin (Illinois Institute of Technology)		
Grant/Contract No.:	NNX09AK65G		
Performance Goal No.:			
Performance Goal Text:			
	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power		

Task Description:	compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably and for example reaching much above 300 W/cm2 for refrigerant R-123.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Progress Report (May 2009 – April 2010) • The designs of two small-scale EHD conduction pumps were finalized. This represents the first time that the EHD conduction pumping concept has been studied with characteristic length scales below 1 mm. • A simple experimental setup was completed that allowed the static pressure head generation of the two EHD conduction pumps to be measured using a pressure transducer (in the absence of net flow generation). • The static pressure generation data previously obtained were then used to give approximate predictions of the flow rate generation. • The pump performance data obtained from this experiment is being written into a paper that will be submitted to the Journal of Electrostatics by May 2010. • An advanced two-phase falling film experiment has been fabricated. A gravity driven flow of liquid falls over a vertically-oriented heater in which are embedded several EHD flushed electrodes. The heater power will be varied up to the critical heat flux level. Use of the EHD electrodes to assist in liquid-wetting of the heated surface is expected to yield significant increases in the critical heat flux level compared to the case with no EHD. An abstract for this concept has been submitted to the 2010 ASME International Mechanical Engineering Congress and Exposition, to be held in November 2010. Data are currently being acquired for the baseline case of no-EHD, and the study of EHD enhancement will commence shortly. • A design of a micro-channel with heat addition has been developed and is ready for fabrication. This project will extend the previous study of single-phase EHD conduction pumping in micro-channels and look at: 1) EHD enhancement of single-phase, internal heat transfer in micro-channels, 2) pumping of two-phase (liquid & vapor) fluids in micro-channels, and 3) EHD enhancement two-phase, internal heat transfer in micro-channels. An abstract for this design has been submitted to the 2010 ASME International Mechanical Engineering Congress and Exposition. • An experimental apparatus to control the dielectric fluid flow distribution in parallel micro tubes with EHD conduction pumping has been designed and currently is being fabricated. The experiment is configured for single phase and two-phase flow with the branch lines acting as evaporators in the latter configuration. An abstract based on this work has been submitted to the 2010 ASME International Mechanical Engineering Congress and Exposition. • The effort at Goddard Space Flight center, directed by Jeffrey Didion, is focused on the development of experimentally validated spaceflight hardware. This year’s accomplishments include: a. Manifest aboard the SubTEC IV Sounding Rocket Mission scheduled for October 2010 at Wallops FLight Facility b. Design and manufacture of EHD pump for sounding rocket flight experiment c. Design of EHD Thermal Experiment for SubTec IV Sounding Rocket Mission d. Design of modular EHD Pump for integration into multi-functional (structuralthermal) cold plate for electronics box cooling
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	Seyed-Yagoobi J, Patel VK, Pearson MR. "Experimental Study of Control of Dielectric Fluid Flow Distribution in Microtubes with EHD Conduction Pumping." 2010 ASME International Mechanical Engineering Congress and Exposition, Vancouver, British Columbia, November 2010. Abstracts, 2010 ASME International Mechanical Engineering Congress and Exposition, November 2010. IMECE2010-38216. , Nov-2010
Abstracts for Journals and Proceedings	Pearson MR, Seyed-Yagoobi J. "Single- and Two-Phase Flow in Microchannels with Heat Transfer and Driven by an EHD Conduction Pump." 2010 ASME International Mechanical Engineering Congress and Exposition, Vancouver, British Columbia, November 2010. Abstracts. 2010 ASME International Mechanical Engineering Congress and Exposition, Vancouver, British Columbia, November 2010. IMECE2010-39228. , Nov-2010
Abstracts for Journals and Proceedings	Pearson MR, Seyed-Yagoobi J. "EHD Conduction-Driven Enhancement of Critical Heat Flux." 2010 ASME International Mechanical Engineering Congress and Exposition, Vancouver, British Columbia, November 2010. Abstracts, 2010 ASME International Mechanical Engineering Congress and Exposition, Vancouver, British Columbia, November 2010. IMECE2010-39227. , Nov-2010

Papers from Meeting Proceedings	Yazdani M, Yagoobi J. "Heat transfer augmentation of parallel flows by means of electric conduction phenomenon in macro- and micro-scale." 2009 ASME Mechanical Engineering Congress and Exposition Lake Buena Vista, Florida, November 13-19, 2009. Proceedings of IMECE2009. 2009 ASME Mechanical Engineering Congress and Exposition, November 13-19, 2009, Lake Buena Vista, Florida, USA. 9 pages. , Nov-2009